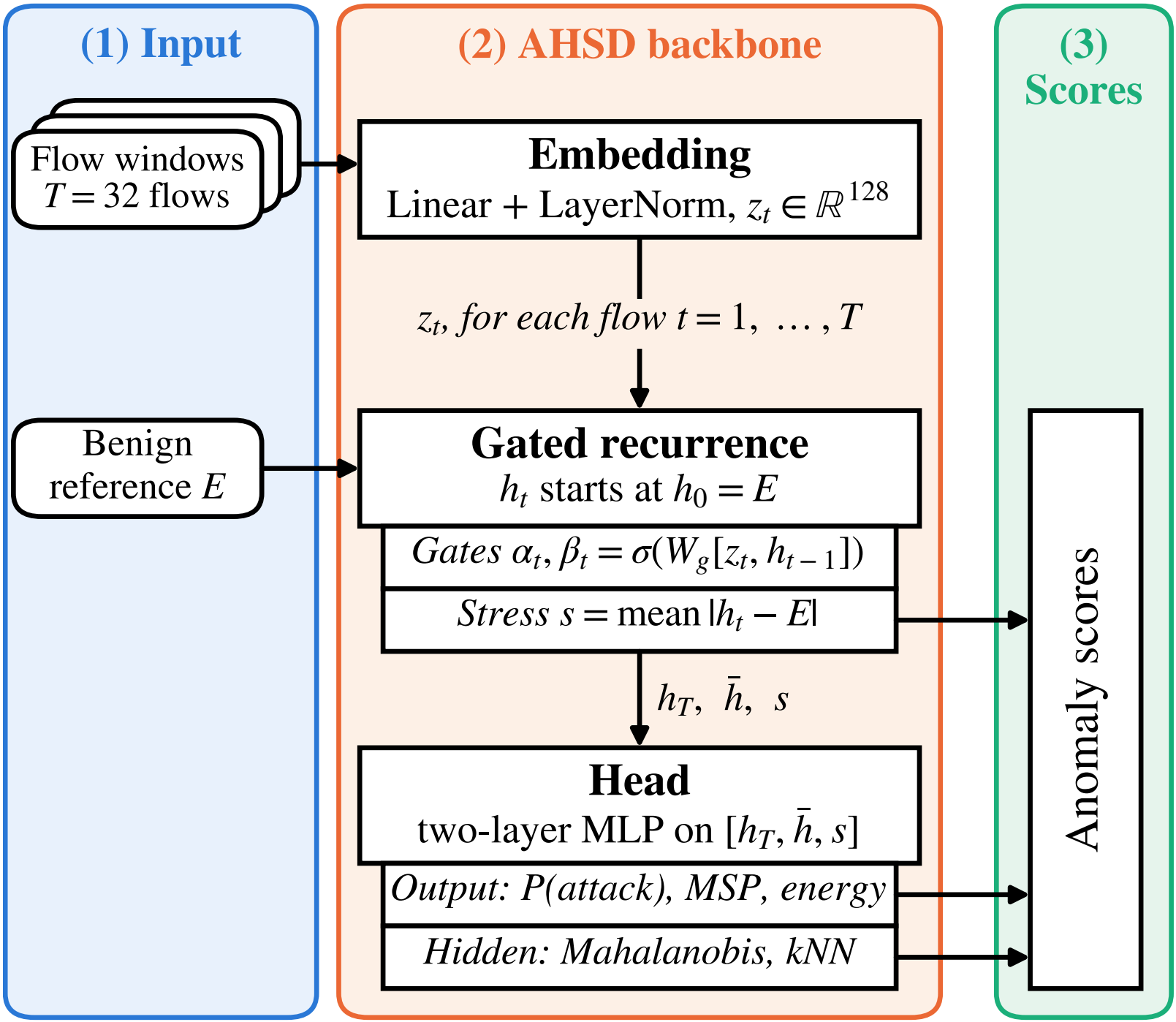
(1) Input
Flow windows
$T = 32$ flows
Benign
reference E
(2) AHSD backbone
Embedding
Linear + LayerNorm, $z_t \in \mathbb{R}^{128}$
z_t, for each flow $t = 1, \ldots, T$
Gated recurrence
h_t starts at $h_0 = E$
Gates $\alpha_t, \beta_t = \sigma(W_g[z_t, h_{t-1}])$
Stress $s = \text{mean}\,|h_t - E|$
$h_T, \ \bar{h}, \ s$
Head
two-layer MLP on $[h_T, \bar{h}, s]$
Output: P(attack), MSP, energy
Hidden: Mahalanobis, kNN
(3) Scores
Anomaly scores